

Session 4 — First light, then deep sky

Practical Astronomy Intensive

9 questions 30 minutes 8 concepts 6 diagnostic

Three of these nine are practical: setting a Dobsonian up, star-hopping to a Messier object with no GoTo, and handing in the log entry for it. The rest manage the expectations that wreck first nights — the Voyager postcard, the Hubble photograph, and the filter that is supposed to brighten a galaxy.

- 1** A pair of binoculars is marked 7×50. What do the two numbers mean, and what should you expect of them?

binoculars Recall Multiple choice

- a 50× magnification in a 7-element optical design — a modest toy to be outgrown once a real telescope arrives
REVEALS '7x50' misread as 50x magnification; binoculars dismissed as a toy
- b 7× magnification and 50 mm objectives, and since 7× is low they can be pushed handheld to 15× or 20× with no loss of steadiness
REVEALS Handheld believed adequate past ~10x, where the observer's own pulse shakes the view
- c 7× magnification and 50 mm objectives — useful for scenery, but they show nothing in the sky a telescope will not show better
REVEALS Binoculars dismissed as a toy to outgrow; their wide field not valued
- d** 7× magnification and 50 mm objectives — the 50 decides how faint you can go, and they will show Jupiter's moons, the Pleiades and hundreds of clusters in a field no telescope can match ✓

The second number is the objective diameter in millimetres — the aperture, and therefore what decides how faint you can go. A 7×50 is the best first instrument and stays many observers' favourite forever. Past about 10× your own pulse shakes the view, and a tripod adapter reveals stars the same glass could not hold still.

- 2** A student arrives at his first dark-sky night having taken his new Dobsonian out of the box that afternoon. He wants to start on M57. What is the order of business?

telescope-operation Apply Multiple choice

- a Fit the highest-power eyepiece in the box first, since M57 is small and needs the magnification to be found
REVEALS Beginner starting at highest power — tiny, dim, fast-drifting field
- b** Finder aligned in daylight, tube out early to cool, lowest-power eyepiece in, centre on something bright, then creep the focuser slowly — and only then go hunting ✓
- c Point it at Lyra and look — a telescope shows you things; that is what it is for
REVEALS Telescope expected to just show you things; finder alignment, cool-down and preparation not seen as necessary
- d Look through it once, and if the view is fuzzy return the scope — a fuzzy view means the optics are bad
REVEALS Fuzzy view blamed on optics when the focuser has not been crept through slowly

Ten minutes of drill separates a magical first night from an hour of hunting empty black sky. Align the finder in daylight on a distant object; let the tube cool; start at lowest power for the widest field; centre something bright; and creep the focuser — focus comes and goes over a hair's width of travel.

- 3 Set up the Dobsonian from its case and get first light on a bright star of your choosing. Work unaided; narrate what you are doing.

telescope-operation

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Assessor watches for, in roughly this order: tube out and cooling before anything else; finder alignment checked (in daylight if the session allows, otherwise on a bright star at low power) and adjusted until finder and eyepiece agree; LOWEST-power eyepiece fitted first, not the shortest one in the case; scope swung onto a bright, easy, unambiguous target; focuser crept through slowly in both directions until the star snaps to a point, rather than racked back and forth in despair. Credit the student who notices a fuzzy star is a focus problem, not an optics problem. Do not credit a fast, confident setup that skips the finder and starts at 200x.

Every part of this is recitable and none of it is reliable until watched. The finder, the cool-down and the lowest-power start are the three things beginners skip, and skipping any one of them turns the night into an hour of empty black sky.

- 4 Using the chart and the finder only, star-hop to a Messier object the assessor names. No GoTo, no app, no help.

star-hopping

Reason

Practical task

ANSWER / WHAT TO WATCH FOR

Assessor watches for: student STARTS at a bright naked-eye anchor star and walks the pattern down to the target one finder field at a time — not by sweeping blindly around where the faint target ought to be; chart turned to face the direction of view; and, critically, that they account for the eyepiece flip — a diagonal mirror-reverses, a Newtonian rotates 180° — rather than hopping confidently in the wrong direction and blaming the chart. Red torch only throughout. Success is landing the object; but credit a student who hops methodically, recognises a wrong turn from the pattern, and recovers, over one who stumbles onto it by luck.

This is the craft skill that turns a chart reader into an observer, and it cannot be examined on paper. Note that GoTo does not make it obsolete: the handset still wants a correct alignment on named stars, and an observer who cannot find Cygnus cannot rescue an alignment that has gone wrong.

- 5 Log the object you have just found, and sketch it. Hand the entry to the assessor before you move the telescope.

observing-log

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Assessor checks the entry carries: date and time, location, instrument and the eyepiece actually in use, seeing and transparency estimates, and — the point — what the student ACTUALLY SAW rather than what the Messier table says is there. The sketch may be a crude pencil smudge; artistic skill is not being marked and should not be mentioned. Credit the student who records the objects they failed to find and the nights of murk alongside the triumphs. Do not accept 'M13 — globular cluster, 25,000 ly' copied from the chart: that is a student who looked at a book, not through an eyepiece.

A log is not bureaucracy for serious people — it is what makes you a better observer, because an object you wrote up is one you actually looked at rather than glanced at. A pencil forces the eye to stay on the target; that is where the detail comes from.

- 6 You split a pretty double in the eyepiece. A student says the two stars must be orbiting one another. Is that safe to assume?

double-variable-stars

Reason

Multiple choice

- a No — many are optical doubles: two stars at wildly different distances that merely happen to line up from where we sit ✓
- b Yes, and any double that will not split just needs more magnification to reveal the pair
REVEALS Splitting doubles believed to be a magnification problem rather than an aperture one
- c It cannot be settled by eye at all — questions like this, and variable-star work, need instruments rather than observers
REVEALS Belief that visual work has no scientific value; eyeball magnitude estimates dismissed
- d Yes — if two stars appear that close together in the eyepiece they must be a bound pair
REVEALS Every double assumed to be a physically bound pair; optical doubles not known

An angular separation is a difference of direction and nothing more — an optical double is a line-of-sight coincidence. Two further points worth making here: splitting a tight pair is set by aperture (the resolving limit), not power; and visual work is genuinely useful — Algol drops over a magnitude every 2.87 days and eyeball estimates against comparison stars remain the standard method.

- 7 Saturn is centred at 120× on a night of mediocre seeing. What gets you the most detail?

planetary-observing

Apply

Multiple choice

- a Take one good look, note it in the log, and move on to the next target while the sky is dark
REVEALS Planet given one glance; the patience the target demands not understood
- b Expect the Voyager view at 120× and, if it is not there, conclude the scope is under-powered
REVEALS Voyager/spacecraft postcard expected at the eyepiece
- c Stay where you are and keep watching — planetary detail surfaces in brief moments of steady air, and you have to be at the eyepiece when they come ✓
- d Reach straight for 300× — Saturn is small and bright, so more power is always the way to more detail on it
REVEALS Highest power reached for on a night whose seeing sets a lower ceiling

Seeing sets the ceiling, and on a boiling night 120× shows far more than 300× ever will. Jupiter at the eyepiece is a small bright disc with two grey belts — the belt structure and the Cassini division only surface after your eye has spent ten patient minutes on it, during the moments the air goes still.

8 A student who has read the Messier table finds M31, looks, and says flatly: 'That's it? A grey smudge.' What is the honest answer?

deepsky-observing

Apply

Multiple choice

- a Nothing is wrong with the view — it is the same at any site, so this is simply what M31 looks like everywhere
REVEALS Sky darkness dismissed as a factor in deep-sky work
- b Turn the power up until the smudge resolves into the photograph's detail
REVEALS Magnification cranked on an extended object, which only spreads the same light and dims it
- c The telescope is too small — a larger instrument would show the spiral arms and colour of the photograph
REVEALS Hubble photograph expected at the eyepiece; the eye's inability to integrate not understood
- d Yes — visually these are faint grey smudges, because a camera integrates photons for hours and the eye cannot integrate over time at all; the skill is learning to see what IS there ✓

This ruins more first nights than anything else, so say it before they look, not after. The photograph stacked hours of photons; the eye integrates over about a tenth of a second and cannot accumulate. Deep sky wants aperture, darkness, LOW power (magnification spreads the same light over more area, dimming an extended object) and dark-adapted eyes.

9 An OIII filter makes the Veil Nebula appear out of nothing. A student immediately screws it in to look at M31. What happens, and why?

visual-filters

Reason

Multiple choice

- a M31 improves enough that the filter is a fair substitute for driving out to a dark site
REVEALS Broadband light-pollution filter treated as a substitute for a dark sky
- b M31 shows more contrast — the filter works on any faint fuzzy, galaxies included
REVEALS Nebula filter expected to rescue a galaxy; the emission-line mechanism not understood
- c M31 gets brighter, as it did for the Veil — that is what a nebula filter does
REVEALS Filter believed to add light rather than subtract it
- d M31 gets dimmer and no easier — a filter subtracts light, passing only the narrow lines an emission nebula radiates, and a galaxy's broad starlight is blocked along with the skyglow ✓

A filter cannot brighten anything — it subtracts. Everything in the field, the nebula included, is dimmer through it; the nebula stands out because the skyglow around it was cut harder. That trick needs the target to radiate in narrow lines, which emission nebulae do and galaxies do not. And a broadband LPR filter trims a couple of lamp wavelengths, not the drive out of town.